

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 3 -- Classifying States | Renewal Processes | Martingales

Duration: 90 minutes. Total: 100 points. Show all working.

Notation: $f(i) = P(\text{ever return to } i \mid \text{start at } i)$. $m(t) = E[N(t)] = \text{renewal function}$. $M(n) = \text{martingale value at time } n$. $T = \text{stopping time}$.

Q1 (30 points). A Markov chain on states $\{0, 1, 2, 3\}$ has transition matrix:

	0	1	2	3
0	[0.5	0.5	0.0	0.0]
1	[0.5	0.0	0.5	0.0]
2	[0.0	0.0	0.3	0.7]
3	[0.0	0.0	0.7	0.3]

- Identify the communicating classes of this chain. Which states communicate with each other? Draw (or describe) the communication structure.
 - Classify each communicating class as recurrent or transient. Use the criterion: a class is recurrent if and only if once you enter it, you never leave with probability 1.
 - For the recurrent class(es), find the stationary distribution restricted to that class. For the transient class, what is the probability that the chain eventually leaves it?
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Q2 (35 points). Machines in a factory fail and are repaired. The time between failures (inter-renewal times) has the distribution:

$$P(X = 2) = 0.4, \quad P(X = 5) = 0.4, \quad P(X = 8) = 0.2$$

Time is measured in days.

- Compute $E[X]$ and $E[X^2]$. By the Elementary Renewal Theorem, what is the long-run rate of failures per day?
 - The factory wants to schedule a preventive maintenance window at the end of day 10. Using $N(10) \sim \text{Poisson}(m(10))$ as an approximation where $m(10) \sim 10/E[X]$, estimate $P(N(10) \geq 5)$ using the Poisson approximation.
 - In stationarity, what is the expected residual time $B(t)$ until the next failure after a random inspection time t ? Use the formula $E[B] = E[X^2] / (2 * E[X])$. Is $E[B]$ greater or less than $E[X]/2$? Explain the Inspection Paradox.
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Q3 (35 points). Let $X(1), X(2), \dots$ be i.i.d. with $P(X(i) = +1) = p$ and $P(X(i) = -1) = q = 1 - p$, and define $S(n) = X(1) + \dots + X(n)$, $S(0) = 0$.

- For $p = q = 1/2$ (symmetric walk), define $M(n) = S(n)^2 - n$. Show $M(n)$ is a martingale

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by computing $E[M(n+1) | S(0), \dots, S(n)]$.

Use: $S(n+1) = S(n) + X(n+1)$, $E[X(n+1)] = 0$, $E[X(n+1)^2] = 1$.

b) For $p = q = 1/2$, let $T = \text{first time } |S(n)| = N \text{ for some fixed } N > 0$ (the walk first exits the interval $(-N, N)$). Apply the OST to $M(n) = S(n)^2 - n$ to find $E[T]$. Note that $S(T)^2 = N^2$ always at stopping.

c) Now let $p \neq 1/2$. Define $M(n) = r^{S(n)}$ where $r = q/p$ (De Moivre's martingale). Verify $E[M(n+1) | F(n)] = M(n)$ by computing $E[r^{X(n+1)}] = p * r^{+1} + q * r^{-1}$ and showing it equals 1.